Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

TECHNICAL CATALOGUE · HAND-DRIVEN · METAL BODY

# CIE/444

Hand-driven insulation tester in anti-magnetic aluminium

## Direct-reading insulation resistance testing

A compact, portable instrument for direct reading of insulation resistance on electrical operating machines, rotating plant equipment, stationary apparatus, insulation control and switching equipment, distribution networks, communication cables, insulators, transformers, industrial installations, cables, electrical components, wiring and other devices capable of carrying the rated voltage. It is a versatile instrument for locating intermittent shorts, defective electrical connections, insulation failure and conductor failure, and for preventing costly repairs and breakdowns.




CIE/444 · ILLUSTRATIVE IMAGE

## 01 HIGHLIGHTS

- ✓ Anti-magnetic aluminium body
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Protected metallic flap over the scale window
- ✓ Long life and strength against vibration and impact
- ✓ Guard terminal and externally accessible infinity adjuster on ranges of 1000 MΩ and above
- ✓ Silent generator
- ✓ Collapsible driven handle, folds back flush into the casing
- ✓ Wide anti-parallax mirror scale (not provided on triple-range and above)
- ✓ Highly sensitive cross-coil indicator
- ✓ Accurate and quick response: 3 seconds
- ✓ Conforms to IS 2992:1987

## 02 TYPICAL APPLICATIONS

- Motors, generators, and other rotating machinery
- Insulation control and switching equipment checks
- Distribution network and communication cable insulation testing
- Transformer and insulator testing; industrial installation checks
- Locating intermittent shorts, defective connections, and developing insulation failures



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## 03 PRINCIPLE AND CONSTRUCTION

### PRINCIPLE

A hand-driven clutch-type A.C. generator and a highly sensitive cross-coil indicator. The generated A.C. voltage is converted to D.C. voltage by silicon diodes.

### CONSTRUCTION

Housed in a sturdy, elegant case of anti-magnetic cast aluminium with a metallic carrying handle. The instrument is dust-free: rubber shoes hold the surface and give it stability, and a protected metallic flap covers the glass window of the scale.

## 04 SALIENT FEATURES

### Silent generator

The A.C. generator runs extremely silently and smoothly because of the excellent dynamic balance of its rotating parts, and is free from mechanical noise.

### Centrifugal governor

A high-grade centrifugal governor holds the voltage constant for accurate measurement. Variation in crank speed does not affect the reading, as a cut-off clutch disengages the drive beyond the rated speed.

### Collapsible handle

The generator driven handle folds back flush into the casing when the tester is not in use.

### Quick response

The response time is 3 seconds, when a resistance equal to the central scale mark is suddenly inserted between the measurement terminals, as per IS 2992:1987.

## 05 PERFORMANCE AND TEST SPECIFICATION

| | |
|---|---|
| Standard | IS 2992:1987 |
| Indicator | High-sensitivity cross-coil movement; wide anti-parallax mirror scale (not provided on triple-range and above instruments) |
| Scale length | 90 mm (approx.) |
| Accuracy | Conforms to IS 2992:1987 |
| Generator speed | 160 r.p.m. or more; the centrifugal governor holds the output voltage constant |
| Response time | 3 seconds |
| High voltage test | Withstands minimum 2000 V or (rated voltage + 1000) V A.C. (rms) between the electrical circuit and the handle for one minute |
| Insulation resistance of the instrument | Better than 50 MΩ at 500 V between the electrical circuit and the containing case |



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## 06 CONSTRUCTION AND PHYSICAL SPECIFICATION

| | |
|---|---|
| Guard terminal and infinity adjuster | **Provided on instruments with an insulation range of 1000 MΩ and above** |
| Body | **Anti-magnetic cast aluminium case with metallic carrying handle and protected metallic flap over the scale window** |
| Weight | **2.5 kg (approx.)** |
| Dimensions | **110 × 128 × 195 mm (approx.)** |

## 07 RANGE OPTIONS BY RATED VOLTAGE

Each instrument is built for **one rated voltage** and **one insulation range**. Choose a rated voltage, then **one** range from its row. Ranges are not combined in one instrument unless one of the dual voltage or dual insulation range options below is selected.

| RATED VOLTAGE | INSULATION RESISTANCE RANGE (MΩ): ANY ONE OF |
|---|---|
| **100 V** | 0–20 OR 0–50 OR 0–100 |
| **250 V** | 0–50 OR 0–100 OR 0–250 |
| **500 V** | 0–100 OR 0–200 OR 0–500 OR 0–1000 OR 0–2000 |
| **1000 V** | 0–100 OR 0–200 OR 0–500 OR 0–1000 OR 0–2000 |
| **2500 V** | 0–1000 OR 0–2000 OR 0–2500 OR 0–5000 OR 0–10,000 OR 0–20,000 OR 50–50,000 |
| **5000 V** | 0–5000 OR 0–10,000 OR 0–20,000 OR 50–50,000 |

*The 100,000 MΩ range at 5000 V is not offered on the metal-body CIE/444 (available on the wooden-body CIE/666).*

## 08 DUAL VOLTAGE AND DUAL INSULATION RANGE OPTIONS

### Dual voltage and dual insulation range

One instrument with two rated voltages, each with its own insulation range, selected by a switch. The selected range must be within the rated resistance listed for that voltage (voltage to measurable insulation range ratio up to 1:2 only; for any other ratio, please check with us).

### Single voltage but dual insulation range

One rated voltage with two insulation ranges. The selected ranges must be within the rated resistance listed for that voltage.

### Continuity range (in 500 V and 1000 V only)

Available as an extra on single-voltage, single-range instruments, or on dual-voltage, dual-range instruments.



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## 09 MULTIRANGE INSULATION TESTER (HAND-DRIVEN): CIE/444/MR

| SERIES | RATED VOLTAGE / INSULATION RANGE (MΩ) | CONTINUITY |
|---|---|---|
| **MR-1** | 1000 V / 0–2000<br>500 V / 0–1000<br>250 V / 0–500 | 0–20 kΩ |
| **MR-2** | 1000 V / 0–200<br>500 V / 0–100<br>250 V / 0–50 | 0–1 kΩ |
| **MR-3** | 500 V / 0–1000<br>250 V / 0–500<br>100 V / 0–200 | 0–5 kΩ |
| **MR-4** | 500 V / 0–100<br>250 V / 0–50<br>100 V / 0–20 | 0–1 kΩ |

*Multirange instruments are provided with a range selector switch, so one instrument covers every voltage and range listed in its series. Anti-parallax mirror scale is not provided on triple-range and above instruments. Any other range can be manufactured against specific requirements.*

## 10 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Carrying case
- Testing leads, 1 m or 2 m (up to 1000 V)
- H.V. testing leads, 2 m (general / special)

## 11 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/444, 500 V / 0–2000 MΩ**. Any range not listed can be manufactured against specific requirements.



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